

T1.1

Scalar first-order bases & direct Adam variants

AdamW

S1 scope: element-wise parameters
S2 signal: raw gradient g_t
S3 state: $m_t = EMA(g_t); v_t = EMA(g_t^2)$
S4 reconstruct: full-size states, no reconstruction
S5 update: decoupled weight decay + adaptive diagonal step

Adam

+ Decouple decay
 S3: EMA 1st/2nd moments
 S5: coupled L2 decay

ADOPT

+ Pre-normalize gradient
 S3/S5: reorder moment update and normalization

AdaBelief

+ Belief variance
 S3: centered belief variance

Adan

+ Add gradient-difference
 S3: gradient-difference memory

Radam

+ Variance rectification
 S5: early adaptive-variance rectification

T1.2

Multi-timescale & momentum correction & variance reduction

AdEMAMix

S1 scope: element-wise parameters
S2 signal: raw gradient g_t
S3 state: fast EMA + slow EMA
S4 reconstruct: identity reconstruction
S5 update: $update \propto \alpha \cdot m_{fast} + (1 - \alpha) \cdot m_{slow}$

Simplified-AdEMAMix

+ Simplify fast-slow mixing
 S3/S5: reduced fast-slow mixing rules

MARS-AdamW

S2: VR-STORM corrected gradient

T1.3

Iterate averaging & automatic tuning

Schedule-Free AdamW

S1 scope: element-wise parameters
S2 signal: gradient at interpolation point
S3 state: maintain $x_t, y_t, z_t; y_t = (1 - c)z_t + c \cdot x_t$
S4 reconstruct: identity reconstruction
S5 update: constant LR + decoupled WD; no LR schedule

DoG

+ online LR-scale estimate
 S5: parameter-free distance step

D-Adaptation

S5: distance lower-bound LR scale

Refined Schedule-Free

+ Refine averaging
 S3/S5: robust iterate averaging

Prodigy

+ improve distance-scale estimate
 S5: online distance-scale estimate